

VARDHAN STEEL & ENERGY LTD.

Coimbatore Works — Unit 3 (Rolling Mill Complex)

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EQUIPMENT INSPECTION & MAINTENANCE REPORT

Report No.:	VSE-MNT-2026-04471	Report Type:	Scheduled Preventive Inspection
Date of Inspection:	18-Jul-2026	Shift:	B Shift (14:00 - 22:00)
Plant / Area:	Rolling Mill Complex, Unit 3	Equipment Tag:	GB-RM3-0207
Equipment Type:	Helical Gearbox (Main Drive Reducer)	Manufacturer:	Elecon Engineering
Model / Serial No.:	MAAG-WPU-450 / SN-88213-C	Installed On:	11-Mar-2016
Inspector:	R. Kannan (Sr. Mechanical Technician)	Employee ID:	EMP-33456
Supervisor Approval:	S. Meenakshi, Reliability Engineer	Approval Date:	19-Jul-2026
Work Order Ref.:	WO-2026-118820	Related P&ID Ref.:	PID-RM3-DRV-014, Rev. C

1. Background & Equipment History

Gearbox GB-RM3-0207 drives the No. 2 stand of the hot rolling mill and has logged approximately 38,400 operating hours since commissioning. The unit underwent a partial bearing replacement in September 2023 (WO-2023-097712) following elevated vibration readings on the output shaft bearing (DE side). Lubrication oil (ISO VG 320 synthetic) was last changed on 02-May-2026, ahead of schedule, due to a minor water ingress event traced to a damaged breather cap during the monsoon shutdown. No major failures have been recorded on this unit in the last 18 months; however, maintenance logs from the previous shift (17-Jul-2026, A Shift) note an intermittent unusual noise reported by the mill operator during start-up, which prompted this unscheduled portion of today's inspection in addition to the routine quarterly check.

2. Inspection Findings Summary

Parameter	Measured Value	Reference Range	Status
Vibration - DE Bearing (mm/s RMS)	4.8	≤ 4.5 (ISO 10816-3, Zone B)	Marginal
Vibration - NDE Bearing (mm/s RMS)	2.1	≤ 4.5	Normal
Oil Temperature (°C)	68	≤ 75	Normal
Casing Surface Temp (°C, IR gun)	71	≤ 80	Normal
Oil Analysis - Fe content (ppm)	142	≤ 100 (caution level)	Above Caution
Oil Analysis - Water content (%)	0.03	≤ 0.05	Normal
Backlash - Output Gear Mesh (mm)	0.31	0.15 - 0.35	Normal
Noise Level (dBA, 1m)	89	≤ 85	Above Limit
Foundation Bolt Torque Check	Pass	As per OEM torque chart	Normal
Coupling Alignment (mm offset)	0.06	≤ 0.08	Normal

Status flags: **Marginal** = trending toward alarm threshold, recheck within 30 days. **Above Caution / Above Limit** = exceeds normal reference band, corrective action recommended.

3. Technician Field Notes

“Checked the gearbox after the noise complaint from A shift. Noise is audible near the output shaft seal, more of a rattling sound during start-up under no load, settles down once loaded. Suspect it could be the seal retainer plate bolts working loose or early-stage bearing wear on the DE side — vibration reading is right at the edge of acceptable so not conclusive on its own. Oil sample sent to lab same day, iron content is a bit high compared to last quarter's 95 ppm reading, worth watching closely. Did not open the inspection cover this round since production could not release the line for more than the scheduled 45-minute window. Recommend a follow-up borescope check and vibration trend review before next quarterly PM, ideally within 2-3 weeks, not urgent enough for an immediate shutdown but should not be pushed past 30 days.”

— R. Kannan, Sr. Mechanical Technician

4. Safety & Compliance Observations

- Lockout-tagout (LOTO) procedure followed correctly per SOP-SAF-0021 prior to inspection; isolation verified with zero-energy check.
- Guard rail near the gearbox platform (north side) has a loose bolt on one panel — logged as a minor housekeeping item, WO-2026-118835 raised for facilities team.
- Gearbox breather cap replacement from May 2026 (post water-ingress event) confirmed intact and correctly seated; no further water ingress signs observed.
- PPE compliance confirmed: hearing protection, safety glasses, and cut-resistant gloves worn throughout the inspection given the 89 dBA noise reading in the immediate vicinity.

5. Recommendations & Follow-up Actions

#	Action	Priority	Target Date	Owner
1	Borescope inspection of output shaft bearing (DE side) and seal retainer bolts	High	05-Aug-2026	Reliability Team
2	Repeat oil analysis (Fe content trend) after 2 weeks of operation	Medium	01-Aug-2026	Lube Lab / R. Kannan
3	Vibration recheck on DE bearing to confirm marginal trend	Medium	08-Aug-2026	Condition Monitoring
4	Repair loose guard rail bolt on inspection platform	Low	25-Aug-2026	Facilities/Civil
5	Update equipment history log in CMMS with this inspection record	Routine	20-Jul-2026	R. Kannan

6. Related Records (Cross-System References)

The following related records exist in separate systems and are referenced here only by ID (not attached) — illustrating the type of cross-document linkage an intelligence platform would need to resolve:

Record Type	System of Record	Reference ID
P&ID Drawing	Engineering Document Vault	PID-RM3-DRV-014, Rev. C
Prior Work Order (bearing replacement)	CMMS (SAP PM)	WO-2023-097712
Lube Oil Analysis Report	Lab LIMS	LAB-OIL-2026-0512
OEM Manual - MAAG WPU-450	Document Management System	OEM-MAN-MAAG-WPU450-R2
Standard Operating Procedure - LOTO	SOP Repository	SOP-SAF-0021
Previous Shift Log (17-Jul-2026, A Shift)	Shift Handover App	SHIFT-LOG-2026-0717-A